

Airline(airlineID, revenue)

Person(personID, occupies[FK9], firstName, lastName)

FK9: occupies -> Location(locID)

On delete, should RESTRICT as people must always occupy a location

On update, should CASCADE since if a loc_id were to change name, people in the location should reflect this change

Flight(flightID, follow[FK11], cost)

FK11: follow -> Route(routeID)

On delete, should RESTRICT as if the plane is already enroute and the route gets canceled it may not know how to get to its final location. If a route needs to be deleted it should have no planes currently following it.

On update, should CASCADE since if a routeID were to change name, the flight following this route should reflect this change.

Route(routeID)

Leg(legID, departure[FK12], arrival[FK13], distance)

FK12: departure -> Airport(airportID)

On delete, should CASCADE since if the airport is taken out of service no legs should be allowed to start or stop there

On update, should CASCADE since if an airport_ID were to change name, the legs traveling to / from that location should reflect this change

FK13: arrival -> Airport(airportID)

On delete, should CASCADE since if the airport is taken out of service no legs should be allowed to start or stop there

On update, should CASCADE since if an airport_ID were to change name, the legs traveling to / from that location should reflect this change

Location(locID)

Airport(airportID, locID[FK6], airportName, city, state, country)

FK6: locID -> Location(locID)

On delete, should be SET NULL as an airport can go out of service and therefore not be given a location ID

On update, should CASCADE since if a loc_id were to change name, the airport in that location should reflect this change

Airplane(tail_num, ownerID[FK1], supports[FK8], locID[FK7], speed, seat_cap)

FK1: ownerID -> Airline(airlineID)

On delete, should CASCADE if an Airline goes out of business we should not track it since we are making this mapping on behalf of the airlines

On update, should CASCADE since an airline could possibly sell its airplanes to another airline or change its name

FK7: locID -> Location(locID)

On delete, should SET NULL since an airplane still should have its info tracked it may just be temporarily out of commission.

On update, should CASCADE since if an airplane's location ID changes it should be reflected in the airplane table

FK8: supports -> Flight(flightID)

On delete, should SET NULL since an airplane does not always have to be supporting a flight

On update, should CASCADE since an airplane's flight it supports should match and reflect all changes made to that flight

Pilot(taxID, personID[FK2], commands[FK10], experience)

FK2: personID -> Person(personID)

On delete, should CASCADE since if a person is deleted from the system we should not keep track of their pilot/passenger information

On update, should RESTRICT since a person's ID should never change.

FK10: commands -> Flight(flightID)

On delete, should SET NULL as a pilot does not always need to be assigned to a plane

On update, should CASCADE to match whatever name change the flight they are commanding had

Passenger(personID[FK3], funds, miles)

FK3: personID -> Person(personID)

On delete, should CASCADE since if a person is deleted from the system we should not keep track of their pilot/passenger information

On update, should RESTRICT since a person's ID should never change.

Boeing((tail_num, ownerID)[FK4], model, maintained)

FK4: tail_num, owner -> Airplane(tail_num, owner)

On delete, should CASCADE since if the airplane gets taken out of commission we should need keep track of its manufacturer information and the same goes if an Airline goes out of business we should not track it since we are making this mapping on behalf of the airlines

On update, should CASCADE since an airline could possibly sell its airplanes to another airline or change its name

Airbus((tail_num, ownerID)[FK5], isNeoVariant)

FK5: tail_num, owner -> Airplane(tail_num, owner)

On delete, should CASCADE since if the airplane gets taken out of commission we should need keep track of its manufacturer information and the same goes if

an Airline goes out of business we should not track it since we are making this mapping on behalf of the airlines

On update, should CASCADE since an airline could possibly sell its airplanes to another airline or change its name

RouteContains(sequence, routeID[FK14], legID[FK15])

FK14: routeID -> Route(routeID)

On delete, should RESTRICT since we need a unique identifier to connect the set of legs and the routeID.

On update, should CASCADE since if a routeID were to change name, the mapping for the set of legs should reflect this change

FK15: legID -> Leg(legID)

On delete, should RESTRICT as deleting a leg while a route is on way may leave a gap on how an airplane is going to end up reaching its final destination.

On update, should CASCADE since if a legID were to change name, the leg contained within contains should reflect this change

License(license type, taxID[FK16])

FK16: taxID -> Pilot(taxID)

On delete, should CASCADE since if the pilot gets fired, we have no need to store their license.

On update, should RESTRICT since a pilot's taxID should never change.

Vacation(location[FK17], personID[FK18], destination, sequence)

FK17: location -> Airport(airportID)

On delete, should CASCADE since if an airport is decommissioned it can no longer be considered a destination to travel to.

On update, should CASCADE since if an airport changes its ID we should reflect this.

FK18: personID -> Passenger(personID)

On delete, should CASCADE since if a person gets removed from the system we no longer should track their vacation

On update, should RESTRICT since a person's ID should never change.